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heatpumps: A heat pump simulation model library and dashboard

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Summary

Heat pumps are a key technology for decarbonizing the heating sector, supplying heat from renewable electricity and low-temperature sources at high efficiency. Their performance and cost strongly depend on the design of the underlying thermodynamic cycle, of which many variants exist. Comparing these variants on a consistent basis requires detailed simulation models that are laborious to build and are often locked away in proprietary software.

`heatpumps` is an open-source Python package that makes such models readily available. It provides a broad collection of ready-to-run heat pump cycle topologies that can be simulated in both design and part-load operation and evaluated with respect to their efficiency, exergy destruction and investment cost on a transparent and comparable basis. The models build on the thermodynamic simulation framework TESPy (Witte & Tuschy, 2020) and the fluid property library CoolProp (Bell et al., 2014), giving access to a wide range of working fluids. All functionality is available through a documented model class API and through an interactive web dashboard, addressing users in research, education, and industry alike.

Statement of need

Many studies in the heating sector emphasize the importance of large electrically driven mechanical compression heat pumps for future energy systems (Paardekooper et al., 2018), (Arpagaus et al., 2018), (Jesper et al., 2021). However, meta analyses indentified gaps in research of the integration and detailed modeling of large-scale heat pumps (Sporleder et al., 2022). Furthermore, a considerable share of studies are limited to simplified Carnot/Lorenz-COP or regression models (Pieper et al., 2020), (Jesper et al., 2021), therefore losing thermodynamic detail, such as cycle topology or component behavior. Studies employing component-based thermodynamic modeling often limit their scope to only one or a few cycle designs (Vannoni et al., 2023).

`heatpumps` contributes to closing this research gap by providing a wide collection of simulation models and analysis features, based on the open-source simulation toolkit TESPy (Witte & Tuschy, 2020). It allows for the steady-state design and part-load simulation of air-to-water and water-to-water heat pumps via a model class API or a Streamlit web-dashboard. Furthermore, `heatpumps` provides the largest selection of cycle topologies available to date, including sub- and transcritical processes, which can be analyzed with the help of state diagrams generated with fluprodia (Witte & Freißmann, 2025). TESPy's integration of CoolProp (Bell et al., 2014) as a fluid backend makes a wide variety of refrigerants available to users. Finally, techno-economic and exergy analyses are provided via a component-based cost estimation approach (Kosmadakis et al., 2020), (Bejan et al., 1996) and the integration of ExerPy (Tomasinelli et al., 2026) respectively.

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In conclusion, heatpumps can broaden the scope of heat pump research while not making compromises in modeling detail. It can support decision-making of planners as well as district heating system operators and engineers of industrial processes. For university education, the provided dashboard can improve accesibility of detailed thermodynamic modeling and strenghten learning with its broad analysis toolbox.

State of the field

Open-source software for modeling heat pumps exists at several levels of abstraction, none of which supports a ready-to-use design-oriented comparison of a broad range of cycle topologies.

At the *system level*, energy system frameworks represent heat pumps in a deliberately reduced manner. `oemof.thermal` (oemof developer group, 2024) derives the COP from the Carnot and a constant second-law efficiency, while data-driven libraries such as `hplib` (Tjaden & Hoops, 2021) fit regression models to certification measurements of market-available units (Jesper et al., 2021). Such models are computationally cheap and well suited to time series simulation and dispatch optimization, but they are agnostic to the thermodynamic cycle itself and can therefore neither represent nor discriminate between process topologies.

At the *component level*, dedicated vapor compression libraries such as `VCLib` (Vering et al., 2022), (RWTH Aachen University, Institute for Energy Efficient Buildings and Indoor Climate (EBC), 2024) and `ACHP` (Bell, 2015) resolve individual compressors and heat exchangers in considerable geometric and empirical detail. Their strength lies in accurately reproducing a specific machine, and they are correspondingly restricted to a small number of flowsheet configurations, predominantly single-stage cycles of residential scale, and provide no economic evaluation.

Between these levels, general-purpose process simulators supply the components from which arbitrary thermal cycles can be assembled. TESPy (Witte & Tuschy, 2020), on which heatpumps is built, and the flowsheeting environment DWSIM (Medeiros, 2024) belong to this category. So do the Modelica libraries AixLib (Müller et al., 2016), Buildings (Wetter et al., 2014) and ThermoCycle (Quoilin et al., 2014), whose emphasis lies on dynamic simulation and control design within building and district energy systems. These tools are general and powerful, but every heat pump topology has to be constructed, parametrized and numerically stabilized by the user. This effort scales with the number of setups to be compared and constitutes a considerable barrier to non-specialist use. The same applies to the proprietary environments frequently employed in heat pump design studies, which additionally limit reproducibility (Wallerand et al., 2018).

heatpumps occupies the gap between these levels by adding a domain-specific layer on top of a general-purpose engine. Each topology is implemented as a model class encapsulating the component and connection network, dynamically generated starting values for numerically unstable parameters, consistency checks against strict thermodynamic limits and post-processing. Three properties follow from this architecture. First, the breadth of the model collection—comprising 72 ready-to-run configurations of simple, intercooler, economizer (open or closed, with serial or parallel compression) and flash tank base cycles, each available as a single-cycle or cascaded and as a sub- or transcritical variant—is to the authors' knowledge unmatched by any other open-source package. It thereby addresses the call for a systematic exploration of the available cycle configurations (Adamson et al., 2022). Second, thermodynamic simulation is coupled to a component-based cost estimation (Kosmadakis et al., 2020), (Bejan et al., 1996) and an exergy analysis (Tomasinelli et al., 2026), so that cycle efficiency and the cost required to achieve it can be weighed against each other within a single tool on a comparable basis. Third, the deliberately superstructure-free formulation keeps every topology explicit and inspectable, and makes the library extensible by contributing further model classes instead of re-encoding a monolithic optimization problem.

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Software design

heatpumps follows a layered, object-oriented design that separates the user interface, the parametrization and the thermodynamic models (Figure 1). The Streamlit dashboard and the Python API both pass a parameter set, derived from one of the bundled JSON presets, to the same model classes. As the dashboard adds no modeling capability of its own, both routes yield identical results, and the same models drive interactive exploration as well as large scripted batch studies without duplication.

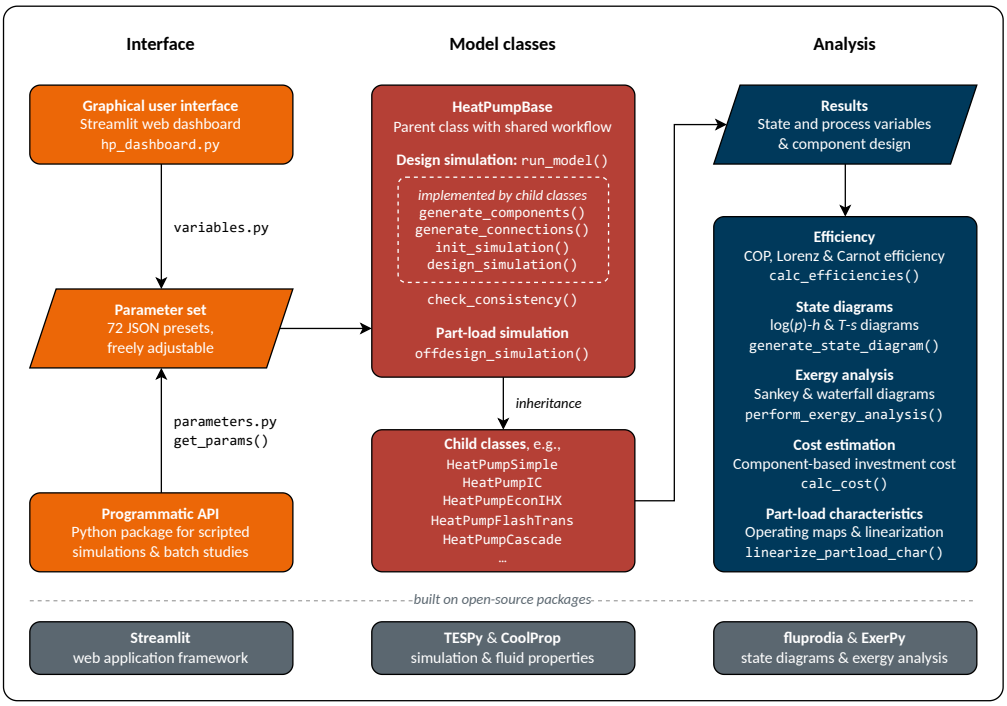


Figure 1: Software architecture of heatpumps. Dashboard and Python API pass a parameter set to the same model classes. The parent class HeatPumpBase implements the shared simulation workflow, whose topology-specific steps (dashed box) are implemented by the child classes. The analysis methods evaluate the solved model. Grey boxes indicate the open-source packages each layer builds on.

At the core, the parent class HeatPumpBase implements everything common to all heat pumps, i.e., the design and part-load simulation workflow, the consistency checks and the analysis methods. Each topology is realized as a child class that only implements the aspects specific to it, namely its components, their connections, and its starting values and design specifications. New topologies can therefore be added by contributing a single child class without touching the shared workflow or the interface. The class names encode the topology, so that for example HeatPumpEconIHX denotes an economizer cycle with an internal heat exchanger and HeatPumpEconIHXTrans its transcritical counterpart.

A recurring challenge in the equation-based simulation of heat pump cycles is the numerical stability of the solver, which strongly depends on the quality of the starting values. heatpumps therefore solves each design in two stages: an initial simulation with robust starting values, generated dynamically from the user's specifications. This first simulation then initializes the final design simulation in which the actual target parameters are imposed. A subsequent consistency check against strict thermodynamic limits flags implausible solutions. The final simulation results can then be explored via a host of analysis methods or serve as the starting point of part-load investigation.

Rather than reimplementing established functionality, heatpumps builds on the surrounding open-source ecosystem (Figure 1). It is distributed via PyPI under the MIT license, tested through an automated test suite and documented on Read the Docs.

Research impact statement

Although heatpumps is a young project, it is already being applied in research. Within the authors' research group it has supported several district heating research projects and provides the modeling basis for a comparative design study of heat pump topologies for district heating systems (Freißmann et al., 2026). Its dashboard was presented at the 2nd Conference of North German Heat Research (Fritz et al., 2025), and the publicly hosted instance on Streamlit Community Cloud has recorded more than 1,000 unique visitors as of October 2026, indicating interest beyond the immediate developer circle.

Since its initial release on PyPI in May 2026, heatpumps has been downloaded over 2,000 times (according to pepy.tech, last checked October 2026). While adoption is still at an early stage, ongoing development is deliberately aimed at broadening the tool's reach: the latest release introduced English-language support to the previously German-only dashboard and expanded the documentation, lowering the barrier to entry for an international audience. Together with the growing research interest in large-scale heat pumps for the decarbonization of the heating sector, this positions heatpumps to make detailed, reproducible heat pump design studies more widely accessible.

AI usage disclosure

The majority of heatpumps was developed without generative AI tools. Only recently (ca. `v1.4.0` onwards) have coding agents (Claude Code, Opus 5 models) been slowly integrated into the developement process. The first draft of this manuscript was written solely by the authors and a second iteration was cleaned up with the help of AI language tools (as above; also: DeepL). All changes were reviewed and verified by the authors, who take full responsibility for the final manuscript.

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